JABATAN KEJURUTERAAN AWAM
SESSION 2:2024/2025
DCC10292 – CONSTRUCTION MATERIALS
(TOPIC 2, 3, 4, & 5)
DURATION: WEEK 1 – WEEK 14

NAME:	**PROGRAMME:**	
REGISTRATION NO.:	**CLASS:**	
ITEM: MINI PROJECT (TOPIC 1, 2, 3, 4, & 5)	**MARKS:** **/ 100**	
LEARNING OUTCOME: Propose selection of material and method in construction practice with consideration of construction specification, condition and sustainability.	**(A3, PLO6)**	
INSTRUCTION:		

You and your team are tasked with a mini project to design a residential building using brick, timber, concrete, and steel. Carefully select and apply these materials based on their specifications, suitability for the building's conditions, and sustainability principles. Provide a thorough explanation for each material choice and detail how they will be used in the construction process.

REQUIREMENT:

1. MATERIAL SELECTION

Explain:

- Figure out the material (brick, concrete, wood and steel) for different parts of the building element/structure of your residential building project.
- Justify the reasons for the selection of your choices based on the criteria as below:

CRITERIA	DESCRIPTION
Cost	Figure out if each material is affordable.
Structural Properties	Check how strong each material is.
Aesthetics	Think about how it looks and the style of the architecture.
Durability	Look at how often it needs to be maintained.
Sustainability	Talk about how long-lasting and eco-friendly each material is, figure out how the building methods affect the environment, point out any environmentally friendly methods or new ideas you used in the building process.

- State the advantages and disadvantages of the chosen construction materials.

2. APPLICATION OF MATERIALS

Discussion:

- Describe the building methods used for each material, such as traditional or advanced method, for example: putting bricks, framing with wood, pouring concrete, and making things out of steel.
- Explain the factors of the method selection.
- Describe the method of application, how these techniques help the building's main structure and how they make it work.

3. GLOBAL PERSPECTIVE

Analysis of Impact: (Choose 1 ONLY)

a) **Cultural**: Think about how the design of your building respects or incorporates cultural aspects from the area.
b) **Social:** Think about the effects on other people, like how it will affect group acceptance and benefits.
c) **Economic**: Look at how the project will affect the local economy, such as by creating jobs and boosting the economy.
d) **Environmental:** Think on the resources used, contribution to carbon emissions, effect to wildlife habitat and biodiversity.
e) **Legal:** Look at the perspective of regulation and compliance, human rights (labour, community) and property rights.

4. SUSTAINABILITY CONSIDERATIONS

- Discuss how your material selections and methods support sustainable construction practices (e.g., reducing waste, energy efficiency, durability).

5. OUTPUT

- Project output can be in the form of AutoCAD /BIM drawing or 3D model.
- The 3D model can be built from any material (sustainable materials are recommended) if it can be used to help explanation during your presentation.

MINI PROJECT PROGRESS

CRITERIA OF PROGRESS	TOPIC	DESCRIPTION
Planning & organization	**Mini Project Progress 1** **Topic 2:** Concrete Materials & Mixture **Topic 3:** Timber **Evaluation : Week 9** **Mini Project Progress 2** **Topic 4:** Brick **Topic 5:** Steel **Evaluation: Week 13** *** Consult with your lecturer for any clarifications or additional guidance throughout the project**.	Make a clear project plan, keep track of your time well, and make sure your team works well together.
Seek information		To help you make project choices, use a variety of reliable sources, make sure the information is useful, and think critically about the data you collect.
Progress documentation		Keep thorough records, regularly report on progress, and be clear in the report.
Materials selection		• Select the key component of the building that require material selection. • Briefly describe the function of each component. • Evaluate each material based on strength and durability, cost, availability and environmental impact. • Justify the materials you chose for each component based on evaluation – why, how, *Sustainability is compulsory. • Discuss pro and cons of the of selected material. **Eg:** Advantages: durable, Disadvantages: costly.
Method of construction (application of materials)		• Explain the specific construction methods that will be used with each selected material. • Discuss any advantages or challenges of working with the material during construction.
Sustainability analysis		Discuss how your material selections and methods support sustainable construction practices. (e.g., reducing waste, energy efficiency, durability).

Submission Format: Your progress report should be structured as follows:

1. **Introduction** (Brief overview of the project)
2. **Material Selection for Each Building Component**
   - Evaluation of options
   - Selection rationale
   - Construction methods
3. **Sustainability Analysis**
4. **Progress Summary and Next Steps**

***** END OF QUESTION *****

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Date: 23.1.2025	Date: 27.1.2025	Date:

Item Assessment Checklist	YES	NO
Consistent with: • AST • Bloom Taxonomy Domain • Scheme Answer	✓	

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Nota: